

Diagnosing Why a Student Is Wrong Under Item Budgets

A Confirmatory Study of Prerequisite-Probe Timing in High-School Chemistry

Manuscript status: confirmatory result binding complete; H1-H4 were evaluated and H5 was excluded pre-outcome with machine-bound evidence. All result values and decisions in the generated sections come from the validated machine contract.

中文摘要

多数自适应诊断系统回答学生“会不会”，本研究进一步追问：在有限题量内，系统能否区分学生“为什么不会”？一化儿将单个化学知识点的状态表示为掌握（M）、前置缺失（P）、思维链不稳（C）和目标知识未学（U）。生产模型中，P与U在本节点二元题上的正确率只差0.10，因此本文不声称结构性、数学性或渐近不可识别，而只研究预先规定的9、15、25题预算下的弱可识别性与实际不可识别性。先导仿真提示：前置探针可显著改善P态诊断，但固定配额插入同类探针可能增加C态误诊。为复证这一现象，确证实验在可信生产题库与未改写的生产推断引擎上比较三种探针时机政策：信念触发的期望信息增益策略、无前置题的固定难度阶梯，以及每三题固定插入一道前置题的阶梯。实验分别报告匹配与有意错设定的反应生成条件；辅助LLM画像实验须同时通过正确率带和目标误概念命中率操纵检验，且不作为真人学习证据。分析计划、种子、停止规则、负结果分支与排除规则均在采集前冻结。确证数字仅由机器生成的结果合同回填；无论先导结论是否复现，均如实报告。

确证结果。H1=partially_supported (complete); H2=not_supported (complete); H3=supported (complete); H4=supported (complete); H5=not evaluated (excluded pre-outcome)。H1 A 臂收敛率=0.128696; 救活效应 95% CI=[0.103478, 0.138261]。H2 伤害效应 95% CI=[0.023478, 0.089565]; 无伤害对照 95% CI=[0.072174, 0.122609]。上述结果仅限于有限题量仿真，不证明真人学习效果或教育效能。

Abstract

Most adaptive diagnostic systems estimate whether a learner can answer a question. This project asks a narrower and harder question: can a short diagnostic distinguish *why* the learner is likely to fail? YHer represents one chemistry concept with four states: mastered (M), prerequisite gap (P), chain instability (C), and unlearned target knowledge (U). On target-local binary questions, the production model assigns P and U correct-response probabilities that differ by only 0.10. The states are asymptotically identifiable under the model; this project therefore makes no claim of structural, mathematical, or asymptotic non-identifiability. It studies **budget-limited weak identifiability**, or **practical non-identifiability at the pre-specified budgets of 9, 15, and 25 items**.

A hypothesis-generating synthetic pilot suggested two simultaneous effects. Fixed prerequisite probes substantially improved P-state diagnosis, but the same fixed quota increased C-state misdiagnosis. The confirmatory protocol therefore compares three timing policies using the trusted production question catalog and unmodified production inference engine: belief-triggered expected information gain (A), a fixed local-question ladder (B), and the same ladder with a prerequisite question every third position (C). Matched and deliberately misspecified response generators are reported separately. A secondary LLM-persona study is gated by accuracy and misconception manipulation checks and is not evidence about human students. The analysis plan, seeds, populations, stopping rules, hypothesis branches, and negative-result policy were frozen before confirmatory responses were generated.

Confirmatory findings. H1=partially_supported (complete); H2=not_supported (complete); H3=supported (complete); H4=supported (complete); H5=not evaluated (excluded pre-outcome)。H1 A rate=0.128696; rescue 95% CI=[0.103478, 0.138261]. H2 harm 95% CI=[0.023478, 0.089565]; no-harm 95% CI=[0.072174, 0.122609]. These are simulated finite-budget results, not evidence of human learning or educational efficacy.

1. Problem and Product Context

YHer is a pre-alpha, local system for a narrow Shanghai high-school chemistry loop: diagnose one concept, route the learner to an existing teacher video or constrained explanation, administer independent practice and held-out questions, and update an event-reconstructable profile. It does not claim human validation, learning gains, retention, or production readiness.

The diagnostic distinction matters operationally. A prerequisite gap calls for an earlier concept; chain instability calls for a worked causal bridge; an unlearned concept calls for a direct introduction. Treating all three as generic incorrectness can route a learner to the wrong intervention.

For a target-local item with normalized difficulty d , the production model uses

$$P(Y = 1 \mid M, P, C, U) = (0.90, \gamma + 0.10, 0.70 - 0.20d, \gamma),$$

where $\gamma = 0.25$ for multiple-choice items and 0.03 for numeric items. A prerequisite item uses

$$P(Y_{pre} = 1 \mid M, P, C, U) = (0.90, \gamma, 0.80, 0.75).$$

The 0.10 P/U local difference is nonzero, so unlimited independent observations can identify the states. Within a short session, however, that evidence may be too weak for correct confident convergence. A prerequisite response creates a larger P/U contrast, but a true C learner is expected to answer prerequisite material correctly with probability 0.80. Asking such probes indiscriminately can therefore move C belief toward the wrong states. The scientific question is about *when* to probe, not whether prerequisite questions are universally beneficial.

2. What Is New and What Is Prior Art

Computerized adaptive testing, adaptive item-selection rules, Bayesian knowledge tracing, and spaced-repetition scheduling are established areas (Frederic M. Lord 1971; David J. Weiss 1982; David J. Weiss and G. Gage Kingsbury 1984; Juan Ramón Barrada et al. 2010; Kyung (Chris) Tyek Han 2018; Albert T. Corbett and John R. Anderson 1994; Radek Pelánek 2017; Behzad Tabibian et al. 2019; Junyao Ye et al. 2022). The production policy uses an expected-information calculation, but this particular entropy calculation is not claimed as a literature contribution. Adaptive selection alone is not this project's contribution.

The contribution is a production-bound test of a specific finite-budget failure mode: whether four diagnostically actionable causes can be separated through a binary response channel, and whether belief-triggered and fixed-quota prerequisite policies produce different rescue and harm profiles. The design also makes model dependence visible through a frozen misspecification condition and a no-repeat common-support sensitivity analysis.

The product includes an **FSRS-inspired** memory projection. FSRS is treated here as software and documentation, not as peer-reviewed validation of YHer's decay formula (Open Spaced Repetition contributors, n.d.; Anki contributors, n.d.). Memory scheduling is outside this one-session confirmatory question.

3. Hypothesis-Generating Pilot

T0 used synthetic difficulty grids, a uniform prior, the then-specified production likelihood constants, a fixed seed, and 4,000 simulated students per cell. It did not use the real trusted-item distribution or the final production selector. Its values are therefore pilot observations only and are never pooled with confirmatory estimates.

Under binary local questions, pilot C correct convergence ranged from 13.9%-34.7% at 9 items, 45.1%-61.3% at 15, and 58.4%-70.1% at 25. At medium difficulty, P correct convergence was 0.0% at both 9 and 15 items. Replacing three of nine local questions with fixed-position prerequisite questions changed P correct convergence from 0.0% to 73.9%, but changed C misdiagnosis from 26.8% to 46.1%. These observations generated the confirmatory hypotheses; they do not support the final claims by themselves.

4. Confirmatory Protocol

Hypotheses

- **H1, P rescue:** at budget 15 under the matched generator, Arm A must achieve at least 0.50 P correct convergence and the paired 95% interval for A minus B must be strictly above zero for full support.
- **H2, timing harm:** at budget 9, the paired 95% interval for C minus A C-state misdiagnosis must be above zero, while the upper bound for A minus B must stay below the +0.05 no-harm margin for full support.
- **H3, adaptive sanity check:** A versus B overall accuracy and convergence time are subordinate because adaptive-testing efficiency is prior art.
- **H4, misspecification:** the H1 rescue and H2 harm point directions are checked under the frozen perturbed generator, with degradation reported.
- **H5, LLM personas:** only predeclared provider completion, accuracy-band, and misconception-hit gates can make the secondary behavior analysis reportable.

The ordered supported, partially-supported, and not-supported branches are specified in `experiments/analysis_plan.md`.

Arms and grid

Arm A calls the production selector at every decision and allows eligible local and prerequisite candidates to compete by expected information gain. Arm B cycles target difficulties 0.25, 0.50, 0.75, and 1.00 and never asks prerequisite questions. Arm C uses the same ladder but replaces positions 3, 6, 9, ..., 24 with prerequisite questions. All observations update the posterior through the production mastery module; no parallel Bayesian updater, EIG rule, or stopping implementation is permitted.

The intention-to-simulate grid contains 27 targets x 4 truth states x 3 arms x 50 replicates x 2 generator conditions, or 32,400 planned journeys. One maximum-25-item trajectory supplies nominal views at 9, 15, and 25. The H1/H2 target population is fixed mechanically before responses by prerequisite-family availability. The primary empirical with-replacement stress estimand discloses item and family repetition. A budget-specific sensitivity analysis uses the same precomputed no-repeat common-support target set for all three arms.

Generator separation, outcomes, and stopping

The matched generator samples from production response probabilities. The misspecified generator independently perturbs per-item slip and guess and adds a bounded replicate-level ability offset; inference keeps the production constants. This stress condition reduces but does not eliminate simulator-inference circularity.

At each nominal budget, the analysis reports correct convergence, converged misdiagnosis, terminal argmax accuracy, four-state confusion matrices, severe M/U errors, convergence time, actual item count, repeat fractions, and paired held-out Brier. Matched held-out Brier is an internal calibration check, not external validation. Confidence intervals use exactly 10,000 target-stratified percentile bootstrap resamples while preserving paired arms.

The production stop rule requires a top-two posterior gap greater than 0.45 and at least three direct local answers. Budget exhaustion is recorded separately from confidence convergence. No interim efficacy look, favorable-cell replacement, post-outcome sample increase, hidden exclusion, or parameter tuning is allowed.

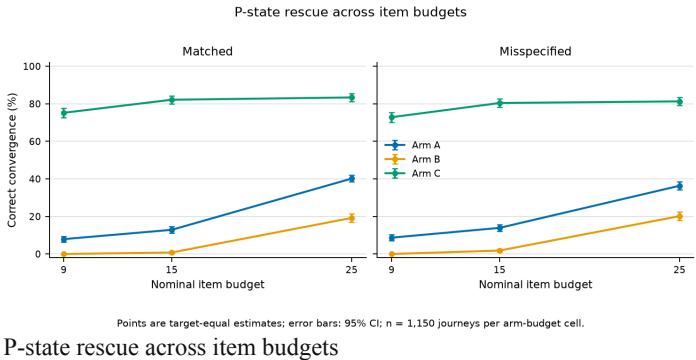
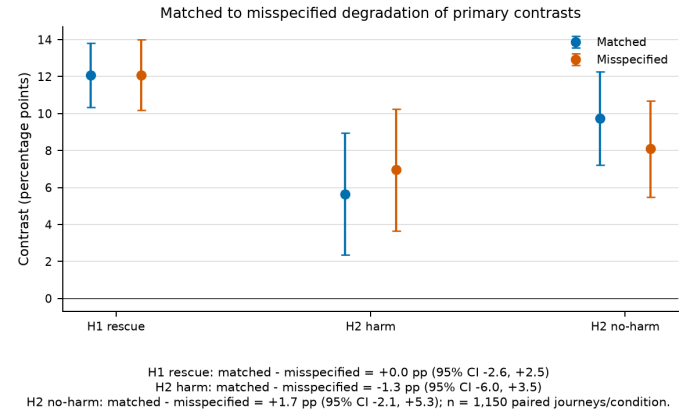
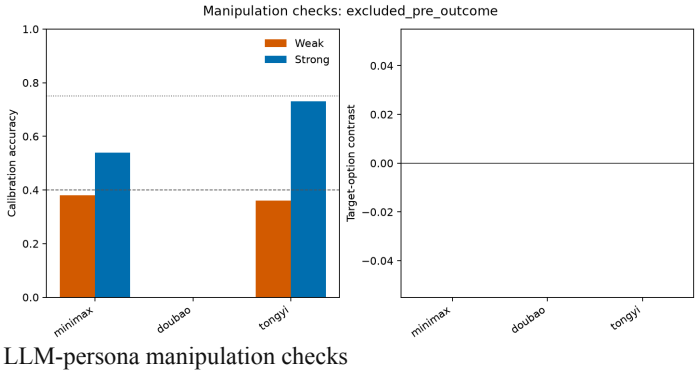
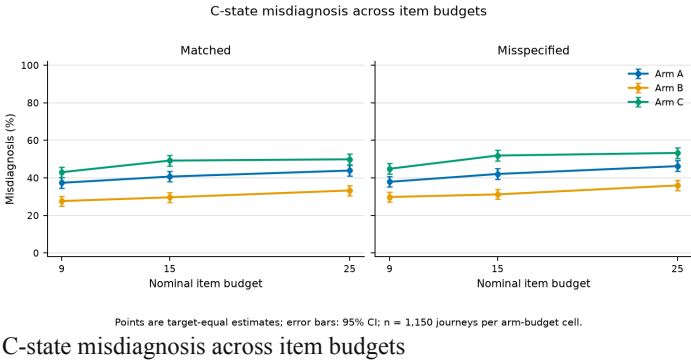
5. Confirmatory Results

This compact table is generated from the same validated contract as the main paper. It reports only decision-driving primary evidence and the mandatory stress/common-support sensitivity evidence.

Machine integrity summary. 30,800 valid of 32,400 intended programmatic journeys; 1,600 predeclared estimand exclusions; H5 excluded pre-outcome with 0 qualifying providers. Provider lifecycle: invalid calibration schema=deepseek/glm/kimi; network interruption=doubao; post calibration exclusion=minimax/tongyi. 12 excluded provider cells and 464 excluded persona cells.

Claim	Decision	Compact machine evidence
H1: A P convergence (complete)	partially_supported	A=12.87%; B=0.78%; A-B=+12.09 pp (95% CI 10.35 to 13.83 pp); no-repeat A-B=+0.50 pp (0.00 to 1.50 pp); stress n=1,150/arm; common-support n=200/arm
H2: C-state misdiagnosis (complete)	not_supported	A=37.39%; B=27.65%; C=43.04%; C-A=+5.65 pp (95% CI 2.35 to 8.96 pp); A-B=+9.74 pp (95% CI 7.22 to 12.26 pp); no-repeat C-A=+2.00 pp (95% CI -3.33 to 7.56 pp); A-B=+13.33 pp (95% CI 9.11 to 17.56 pp); stress n=1,150/arm; common-support n=450/arm
H3: A-B terminal accuracy (complete)	supported	+10.04 pp (95% CI 8.89 to 11.19 pp); median convergence A=8, B=11 items
H4: misspecified rescue and harm (complete)	supported	rescue=+12.09 pp (95% CI 10.17 to 14.00 pp); fixed-probe harm=+6.96 pp (95% CI 3.65 to 10.26 pp)
H5: LLM personas (excluded pre-outcome)	not evaluated	excluded pre-outcome; 0 qualifying providers; no outcome decision

Selected verified figures



Matched-to-misspecified contrast degradation

Discussion

- H1.** The frozen P-state rescue criterion was partially supported. Some required evidence favored rescue, but the full conjunction did not pass; the paper therefore makes no full-support claim.
- H2.** The criterion was not supported. The fixed-quota harm contrast remained positive, but the belief-triggered arm failed the no-harm requirement relative to the local-only arm; this is not a reversal of the fixed-quota pattern.
- H3.** The subordinate adaptive sanity check was supported. It remains secondary because adaptive-testing efficiency is established prior art.
- H4.** Under the frozen misspecification stress, both predicted directions persisted. This result describes sensitivity to one declared perturbation family and cannot establish behavioral realism.
- H5.** The LLM-persona hypothesis was excluded pre-outcome because its machine mapping gate was unavailable. It was not evaluated and is not reported as a negative finding.

These findings remain limited to simulated, finite-budget diagnosis in the tested catalog. They do not establish structural non-identifiability, human validity, learning gains, or educational efficacy.

6. Responsibility, Limitations, and Reproducibility

The project uses three responsibility layers. The student set direction and requirements and defined the pre-outcome quality and honesty gates documented in `decision_log.md`. AI systems proposed designs, implemented automation, executed the simulation and analysis, and drafted the current interpretation and prose under those gates. Commits, an experiment tag, hashes, seeds, tests, provenance envelopes, and isolation attestations form the audit layer. Final interpretation and any submission decision remain pending student review.

The main limitations are explicit: simulated learners are not humans; matched simulation is partly circular; the misspecification family is not exhaustive; the truth state is static; evidence is binary and multiple-choice-heavy; only a subset of the 135-node graph is open; sparse pools may repeat; held-out Brier is internal; and the work comes from one project with extensive AI assistance. Prior LLM-simulated-student work motivates explicit validation and limitation analysis (Xinyi Lu and Xu Wang 2024; Zhengyuan Liu et al. 2024; Hyoungwook Jin et al. 2025; Tao Wu et al. 2025; Alexander Scarlatos et al. 2025, 2026). In this project, H5 is secondary and manipulation-gated; that status is an internal design fact.

The analysis plan, canonical config, seed derivation, source hashes, and isolation rules are repository-bound. Raw confirmatory artifacts must remain in the isolated simulation store and cannot update real profiles or logs. This draft does not publish, push, submit, or mint a DOI.

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